Assignment
Tutorial 2+ Exponential distribution

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Submitted to-

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"Exponential Distribution"

The exponential distribution has density function,

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$
$$= 0 \qquad \text{otherwise}$$

where the parameter λ is real, positive constant. The graph of exponential density is-

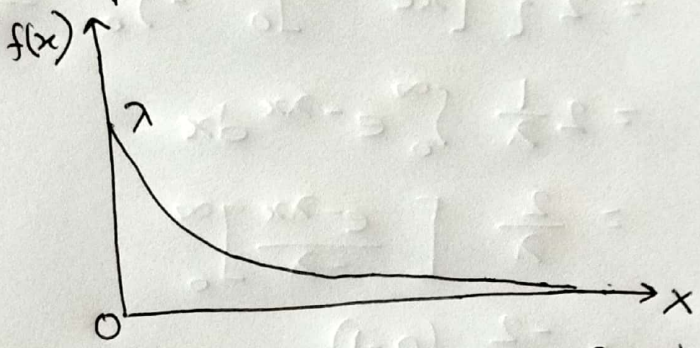


fig- Exponential density function.

The Mean, Variance, Moment-generating function and CDF of exponential function —

Mean, $E = \int_0^\infty x \lambda e^{-\lambda x} dx$

$$= \lambda \left\{ \left[\frac{x e^{-\lambda x}}{-\lambda} \right]_0^\infty - \int_0^\infty \frac{e^{-\lambda x}}{-\lambda} dx \right\}$$

$$= \lambda \frac{1}{\lambda} \int_0^\infty e^{-\lambda x} dx$$

$$= \left[\frac{e^{-\lambda x}}{-\lambda} \right]_0^\infty$$

$$= -\frac{1}{\lambda}(0-1) = \frac{1}{\lambda}.$$

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Variance

$$V(x) = E(x^2) - E(x)^2$$

$$E(x^2) = \int_0^\infty x^2 \lambda e^{-\lambda x} dx$$

$$= \lambda \left\{ \left[x^2 \frac{e^{-\lambda x}}{-\lambda} \right]_0^\infty - \int_0^\infty 2x \frac{1}{-\lambda} e^{-\lambda x} dx \right\}$$

$$= \lambda \frac{2}{\lambda} \int_0^\infty x e^{-\lambda x} dx$$

$$= 2 \left\{ \left[x e^{-\lambda x} \right]_0^\infty - \int_0^\infty \frac{e^{-\lambda x}}{-\lambda} dx \right\}$$

$$= 2 \frac{1}{\lambda} \int_0^\infty e^{-\lambda x} dx$$

$$= \frac{2}{\lambda} \left[\frac{e^{-\lambda x}}{-\lambda} \right]_0^\infty$$

$$= \frac{-2}{\lambda^2} (0 - 1)$$

$$= \frac{2}{\lambda^2}$$

$$\text{Var}(x) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

$$M_x(t) = \int_0^\infty e^{tx} \lambda e^{-x\lambda} dx$$

$$= \lambda \int_0^\infty e^{-x(\lambda - t)} dx$$

$$= \lambda \left[\frac{e^{-x(\lambda-t)}}{-(\lambda-t)} \right]_0^\infty$$

$$= \frac{\lambda}{-(\lambda-t)} (0 - 1)$$

$$= \frac{\lambda}{\lambda(1 - \frac{t}{\lambda})} = (1 - \frac{t}{\lambda})^{-1}$$

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CDF, $$F(x) = \int_0^x \lambda e^{-\lambda t}\, dt$$

$$= \lambda \left[\frac{e^{-t\lambda}}{-\lambda}\right]_0^x$$

$$= -1\left(e^{-x\lambda} - e^{0}\right)$$

$$= 1 - e^{-\lambda x}$$

$$\therefore F(x) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

"Application"

⇒ Measure expected time for an event to occur.

⇒ The exponential Distribution is often useful in situations involving waiting times.

⇒ Some situations involving an exponential random variable-

1. The time it takes for your message to arrive
2. Distance between mutations on DNA strand
3. Times between calls to a help desk.
5. Lifetime.

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"Example"

A electronic component is known to have a life represented by an exponential density with failure rate 10^5 failures per hour ($\lambda = 10^{-5}$). What is the fraction of components that would fail before mean life?

Here, $\lambda = 10^{-5}$

$$E = \frac{1}{\lambda} = 10^5$$

$$P\left(T \leq \frac{1}{\lambda}\right) = \int_0^{\frac{1}{\lambda}} \lambda e^{-\lambda x} dx$$

$$= \lambda \int_0^{\frac{1}{\lambda}} e^{-\lambda x} dx$$

$$= \lambda \left[\frac{e^{-\lambda x}}{-\lambda} \right]_0^{\frac{1}{\lambda}}$$

$$= -\left(e^{-1} - 1\right)$$

$$= 1 - e^{-1}$$

$$= 0.63212$$

$$= 63.212\%$$

So, 63.212 % of terms would fail before 10^5 times.

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Tutorial-02

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Q1. Define Gamma Distribution and Weibull distribution. Write down some application of Gamma Distribution?

For $n>0$ the gamma function is

$$\Gamma n = \int_0^\infty x^{n-1} e^{-x} dx$$

A continuous variable x is said to have a gamma distribution if the pdf of x is

$$f(x) = \frac{\lambda}{\Gamma r} (\lambda x)^{r-1} e^{-\lambda x}, \quad x>0$$

where parameter $r>0$, $\lambda>0$

r is called the shape parameter

λ is called the scale parameter

A random variable x is said to have a weibull distribution if the density function is-

$$f(x) = \frac{\beta}{\delta}\left(\frac{x-\gamma}{\delta}\right)^{\beta-1} \exp\left(-\left(\frac{x-\gamma}{\delta}\right)^{\beta}\right); \quad x \geq \beta$$

$$= 0 \quad ; \text{otherwise}$$

Its parameters - γ is $(-\infty < \gamma < \infty)$ is location parameter,

$\delta>0$ scale parameter

$\beta>0$ shape parameter.

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Application of Gamma Distribution

- Modeling size of insurance claims, rainfall
- Predicting the wait time until future events.
- Wait time modeling
- Reliability modeling
- Service time modeling

Q2. What do you mean by Binomial ~~vib~~ variate? Suppose X follows Binomial distribution with parameters n and p then show that

(i) $E\left(\frac{X}{n}-P\right)^2 = \frac{pq}{n}$ (ii) $cov\left(\frac{x}{n}, \frac{n-x}{n}\right) = \frac{-pq}{n}$

A binomial distribution can be thought of as simply the probability of a SUCCESS or ~~FAILURE~~ FAILURE outcome in an experiment or survey that is repeated multiple time. Binomial distribution has two possible outcomes (bi means two). For example a coin toss has only two outcomes: head or tails.

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(i) $E\left(\frac{x}{n}-p\right)^2$

$= E\left(\frac{x^2}{n^2} - \frac{2px}{n} + p^2\right)$

$= \frac{E(x^2)}{n^2} - \frac{2p}{n}E(x) + E(p^2)$

$= \frac{1}{n^2}\{n(n-1)p^2+np\} - \frac{2p}{n}\cdot np + p^2$

$\left[\because E(a) = a;\ E(x) = np,\ E(x^2) = n(n-1)p^2+np\right]$

$= \frac{1}{n^2}\left(n^2p^2 - np^2 + np\right) - p^2$

$= p^2 - \frac{p^2}{n} + \frac{p}{n} - p^2$

$= \frac{p(1-p)}{n}$

$= \frac{pq}{n}\ [p+q=1]$

= R.H.S (showed)

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(ii) we know

$cov(X,Y) = E(XY) - E(X)E(Y)$

$\therefore cov\left(\frac{x}{n}, \frac{n-x}{n}\right)$

$= E\left(\frac{x}{n}\cdot\frac{n-x}{n}\right) - E\left(\frac{x}{n}\right)E\left(\frac{n-x}{n}\right)$

$= \frac{1}{n^2}E(nx-x^2) - \frac{E(x)}{n}\cdot\frac{1}{n}\{E(n) - E(x)\}$

$= \frac{1}{n^2}\left(n\cdot np - n(n-1)p^2 + np\right) - \frac{np}{n^2}\left(-np+n\right)$

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$$= \frac{1}{n^2}\left(n^2p - n^2p^2 + np^2 + np\right) = \frac{p}{n}(np + n)$$

$$= p - p^2 + \frac{p^2}{n} - \frac{p}{n} + p^2 - p$$

$$= \frac{p(p-1)}{n}$$

$$= \frac{-pq}{n}$$

= R.H.S (proved)

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Q3. A particular telephone number is used to receive both voice calls and fax messages. Suppose that 30% of the incoming calls involve fax message and consider a sample of 40 incoming calls. What is the probability that

(a) At most six of the calls involve a fax massages?

(b) Exactly six of the calls involve a fax msg?

(c) At least six of the calls involve a fax msg?

(d) More than six of the calls involve a fax msg?

Here $n = 40$

Let $p = 30\% = 0.30$

$q = 1 - 0.3 = 0.7$

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(a) $P(X \leq 6) = \sum_{x=0}^{6} f(x)$

$= \sum_{x=0}^{6} \binom{n}{x} p^x q^{n-x}$

$= \sum_{x=0}^{6} \binom{40}{x} (0.3)^x (0.7)^{40-x}$

$= 0.0237$

(b) $P(X=6) = \binom{n}{6} (0.3)^6 (0.7)^{40-6}$

$= 0.01514$

(c) $P(X \geq 6) = 1 - P(X < 6)$

$= 1 - \{P(X \leq 6) - P(X=6)\}$

$= 1 - (0.0237 - 0.1514)$

$= 1 - 0.00856$

$= 0.9914$

(d) $P(X > 6) = 1 - P(X \leq 6)$

$= 1 - 0.0237$

$= 0.9763.$

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Q4. Consider writing into a computer disk and then sending it through a certifier that counts the number of missing pulses. Suppose the number X has a poisson distribution with parameter $\lambda = 0.2$

(a) what is the probability that the disk has exactly one missing pulse?

(b) what is the probability that the disk has at least two missing pulses?

(c) If two disks are independently selected, what is the probability neither contains a missing pulse?

Suppose, A number X has a poisson distribution with $\lambda = 0.2$

(a) the mass function

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!} \qquad x = 0, 1, 2 - - -$$

So, the probability that the disk has exactly one missing pulse:

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$$P(x=1) = \frac{e^{-\lambda}\lambda^x}{x!}$$

$$= \frac{e^{-0.2}(0.2)^1}{1!}$$

$$= 0.1637$$

(b) What is the probability that the disk has at least two missing pulses?

$$P(X \geq 2) = 1 - P(X \leq 1)$$

$$= 1 - [P(x=0) + P(x=1)]$$

$$= 1 - \left[\frac{e^{-0.2} + \lambda^0}{0!} + 0.1637\right]$$

$$= 1 - [0.9824]$$

$$= 0.01757$$

(c) If two disks are independently selected, what is the probability that neither contains a missing pulse?

Let A → first disk has no missing pulse,

B → second disk has no missing pulse.

So, $P(A) = P(x=0) = \frac{e^{-0.2}\lambda^0}{0!} = .8187$

$P(B) = P(X=0) = 0.8187$

$\therefore P(A \cap B) = P(A).P(B)$ as they are independently selected.

$P(A \cap B) = 0.8187 \times 0.8187$

$= 0.67026$ (Ans).

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Qs. Find Mean, Varriance and moment generating function of Exponential distribution,

Exponential distribution function,

$$f(x;\lambda) = \lambda e^{-\lambda x} ; \quad x>0$$

Mean, $E = \int_0^\infty x f(x)\,dx$

$$= \int_0^\infty x\,\lambda e^{-\lambda x}\,dx$$

$$= \lambda\left\{\left[x\frac{e^{-\lambda x}}{-\lambda}\right]_0^\infty - \int_0^\infty \frac{e^{-\lambda x}}{-\lambda}\,dx\right\}$$

$$= \lambda\left\{0+0+\frac{1}{\lambda}\int_0^\infty e^{-\lambda x}\,dx\right\}$$

$$= \left[\frac{e^{-\lambda x}}{-\lambda}\right]_0^\infty$$

$$= \frac{-1}{\lambda}[0-1]$$

$$= \frac{1}{\lambda}$$

Varriance,

$$V(x) = E(x^2) - \{E(x)\}^2$$

$$= \int_0^\infty x^2\,\lambda e^{-\lambda x}\,dx - \frac{1}{\lambda^2}$$

$$E(x^2) = \int_0^\infty \lambda\, x^2 e^{-\lambda x}\,dx$$

$$= \lambda\left\{\left[\frac{x^2 e^{-\lambda x}}{-\lambda}\right]_0^\infty - \int_0^\infty 2x\frac{e^{-\lambda x}}{-\lambda}\,dx\right\}$$

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$$= \lambda \left\{ \frac{2}{\lambda} \int_0^\infty x e^{-\lambda x} dx \right\}$$

$$= 2 \left\{ \left[\frac{x e^{-\lambda x}}{-\lambda} \right]_0^\infty - \int_0^\infty \frac{e^{-\lambda x}}{-\lambda} dx \right\}$$

$$= 2 \frac{1}{\lambda} \left[\frac{e^{-\lambda x}}{-\lambda} \right]_0^\infty$$

$$= \frac{-2}{\lambda^2} (0 - 1)$$

$$= \frac{2}{\lambda^2}$$

$$V(x) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2}$$

$$= \frac{1}{\lambda^2}$$

$$M_x(t) = E(e^{tx})$$

$$= \int_0^\infty e^{tx} \lambda e^{-x\lambda} dx$$

$$= \lambda \int_0^\infty e^{-x(\lambda - t)} dx$$

$$= \lambda \left[\frac{e^{-x(\lambda-t)}}{-(\lambda-t)} \right]_0^\infty$$

$$= \frac{-\lambda}{\lambda - t} (-1)$$

$$= \frac{\lambda}{\lambda(1 - \frac{t}{\lambda})}$$

$$= (1 - \frac{t}{\lambda})^{-1}$$

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